

3.3.1

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Question

The points $(3, 4)$ and $(2, -6)$ are situated on the _____ of the line $3x - 4y - 8 = 0$.

Theoretical Solution

Let us solve the given equation theoretically and then verify the solution computationally

According to the question,

Given two points

$$\mathbf{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 2 \\ -6 \end{pmatrix} \quad (1)$$

$$\mathbf{n} = \begin{pmatrix} -3 \\ 4 \\ 1 \end{pmatrix} \quad (2)$$

Equation of line is

$$\mathbf{n}^T \mathbf{x} = 1 \quad (3)$$

Theoretical Solution

$$\mathbf{n}^T \mathbf{a} - 1 = \frac{3}{4} \quad (4)$$

$$\mathbf{n}^T \mathbf{b} - 1 = \frac{-17}{2} \quad (5)$$

As the values are in opposite sign the points are situated in opposite direction w.r.t line

C Code

```
#include <stdio.h>

int main() {
    // Line:  $3x - 4y - 8 = 0$ 
    // Points
    int x1 = 3, y1 = 4;
    int x2 = 2, y2 = -6;

    // Evaluate  $f(x,y) = 3x - 4y - 8$ 
    int f1 = 3*x1 - 4*y1 - 8;
    int f2 = 3*x2 - 4*y2 - 8;

    printf("Line equation:  $3x - 4y - 8 = 0$ \n");

    printf("For point (3,4),  $f(x,y) = %d \rightarrow$  ", f1);
    if(f1 == 0) printf("Point lies on the line.\n");
    else if(f1 > 0) printf("Point lies on one side of the line.\n");
    else printf("Point lies on the other side of the line.\n");
}
```

C Code - Resultant velocity

```
printf("For point (2,-6), f(x,y) = %d -> ", f2);
if(f2 == 0) printf("Point lies on the line.\n");
else if(f2 > 0) printf("Point lies on the same side as (3,4)
    .\n");
else printf("Point lies on the opposite side of (3,4).\n");

// Final answer
if ((f1 > 0 && f2 > 0) || (f1 < 0 && f2 < 0)) {
    printf("\nBoth points are on the SAME side of the line.\n
        ");
} else if (f1 == 0 || f2 == 0) {
    printf("\nOne of the points lies on the line.\n");
} else {
    printf("\nThe points are on OPPOSITE sides of the line.\n
        ");
}

return 0;
}
```

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

# Define the line  $3x - 4y - 8 = 0 \Rightarrow y = (3x - 8)/4$ 
x = np.linspace(-2, 6, 400)
y = (3*x - 8)/4

# Points
points = [(3, 4), (2, -6)]

# Plot line
plt.plot(x, y, 'b', label='3x - 4y - 8 = 0')

# Plot points
for px, py in points:
    plt.plot(px, py, 'ro')
    plt.text(px+0.2, py, f"({px},{py})", fontsize=10)
```

```
# Formatting
plt.axhline(0, color='black', linewidth=0.5)
plt.axvline(0, color='black', linewidth=0.5)
plt.grid(True, linestyle='--', alpha=0.6)
plt.xlabel("x")
plt.ylabel("y")
plt.legend()
plt.title("Line and Points")

# Save figure
plt.savefig("line_points.png", dpi=300)
plt.show()
```


